Foundational Verification of Probabilistic Programs

Mechanising the Lilac probabilistic separation logic in Lean

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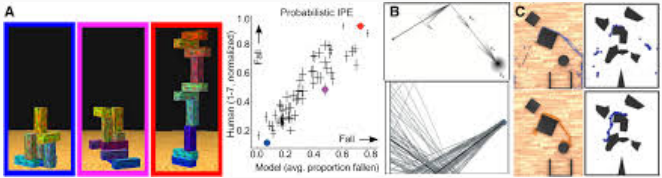
Probabilistic Programming is Everywhere

Probabilistic programming is everywhere

The most general feature of a PPL — sampling from a distribution — lives in the standard library of every major language.

- **Cryptography** — protocols, ZK proofs
- **Differential privacy**
- **Randomised algorithms**
- **Distributed systems**
- **Bayesian inference** & probabilistic ML
- **Scientific simulation** — climate, physics
- **Robotics & perception** — intuitive physics



Discrete vs. continuous

The dividing line that matters runs through the applications.

Discrete

Coin flips, Bernoulli / categorical choices, dice.

Cryptography, differential privacy, randomised algorithms — Applications traditionally within CS - much larger focus.

Continuous

Gaussian noise, sensor models, Bayesian priors over $\mathbb{R}$, physical quantities.

Scientific simulation, Bayesian inference, ML — where the real-world modelling happens.

Continuous distributions are where prior verification work fell short — and where this project makes its **first mechanised** contribution.

Verifying probabilistic programs is notoriously hard

- Conventional testing fails: a rare *correct* outlier execution and a genuine bug look *identical*.
- Correctness is a property of the whole distribution, not of any single run.
- Reasoning about independence and conditioning is subtle and error-prone by hand.

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So we need proof

Machine-checked, **foundational** arguments about distributions — resting on nothing but the axioms of mathematics.

Contribution

The first mechanisation of foundational verification

for probabilistic programs

supporting continuous distributions.

Contribution

1. Soundness proofs
2. Interactive *Proof Mode* using Iris

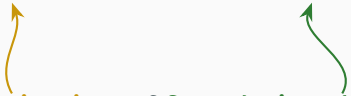
The first mechanisation of foundational verification for probabilistic programs supporting continuous distributions.

Contribution

1. Soundness proofs
2. Interactive *Proof Mode*

using Iris

Using a theorem prover
(Lean/Rocq/Agda)
Assuming only axioms of math[1]



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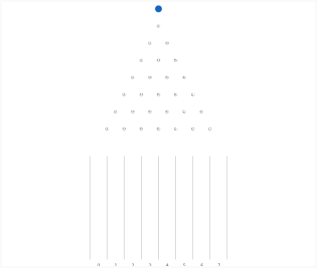
[1] Background Sec 2.4.1 “Dependent Type Theory”

A program is a distribution over runs

$$\text{flip} : (m : \mathbb{R}) \to \text{Distrib}\ \{m, m+1\}$$

an individual "run" of the program



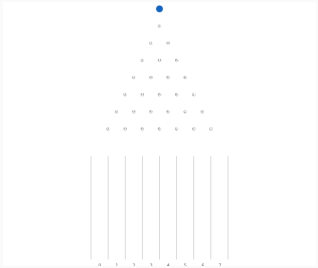
Honing in on a single marble

$$X_0 \leftarrow \text{flip } 0$$
$$X_1 \leftarrow \text{flip } X_0$$
$$X_2 \leftarrow \text{flip } X_1$$
$$\text{ret } X_2$$

$$\Big|\Big|\Big|$$

$$\text{for } (3,\ \text{flip } 0,\ \lambda X.\ \text{flip } X)$$

The program as a whole



All the marbles at once

Contributions, “revised”

Now that the motivation is in place: what did we mechanise, and why does it matter?

- **Lilac** is a probabilistic separation logic whose guiding idea is

  $$\text{“}\textit{separation} = \textit{independence of random variables}\text{”}$$

  The separating conjunction $P * Q$ asserts that the random variables described by P and Q are probabilistically independent — and Lilac supports **continuous** distributions.
- We give the **first mechanisation of Core[2] Lilac in Lean 4**, the first mechanisation of *any* probabilistic separation logic in Lean.

[2] Extended Lilac adds reasoning about conditioning

What was mechanised

1. **Soundness, from the ground up.** Core Lilac is built as a model over probability spaces with finite footprint, and every program-logic proof rule is proved sound against that model:

`wp_ret` `wp_bind` `wp_unif` `wp_cons` `wp_frame`

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2. **An interactive proof interface.** Lilac is instantiated as a **bunched-implication logic** on the Lean port of Iris, so proofs are carried out in the *Iris proof mode*.
3. **Worked examples.** Specifications of sampling programs proved in that proof mode — including constructing a separating conjunct of two independent uniform samples.